Difference between ionization and Dissociation:

Ionization	Dissociation
1. It is the process which produces new Charged particles.	It is the process of separation of charges particles which already exists in a compound.
2. It involves polar covalent compounds or metals.	It involves polar ionic compounds .
3. Irreversible	Reversible
4. Involves covalent bond between atom.	Involves ionic bond between atom.
5. Always produces charged particles.	It produces either charged particles or electrically neutral praticles.

Molecularity and Order

Molecularity	Order
1. Molecularity of a reaction is the number of atoms, ions or molecules taking part in a reaction as in chemical equation.	Order of reaction is the total power of concentration of reactants, which is proportional to the experimental facts.
2. Molecularity is a theoretical concept which depends on chemical equation partly.	Order of reaction is an experimental fact which depends on experiment, observation and conclusion.
3. Molecularity can be fracton	Order of reaction never be fraction
4. Example: • $2\ NH_3\ (g) \longrightarrow 3\ H_2\ (g) + N_2\ (g)$ Molecularity = 2	Example: $2\ NH_3\ (g) \longrightarrow 3\ H_2\ (g) + N_2\ (g)$ Order- Zero order reaction

Physical Absorption and Chemical Absorption

Physical Absorption	Chemical Absorption
1. reversible.	irreversible
2. Balance is easily achieved	The balance is done gradually
3. Due to intermolecular van der Waal forces, physical absorption occurs	Due to the formation of chemical bonds, chemical absorption occurs
4. Low heat of adsorption	The heat of adsorption is very large
5. Decreases with increasing temperature	Increases with increasing temperature